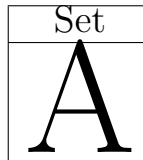


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Brac University
Semester: Fall 2025
Course Code: CSE430
Digital Signal Processing
Faculty: EHQ



Assessment: *Quiz 4*
Duration: 25 minutes
Date: December 29, 2025
Full Marks : 20

$$e^{\pm j\omega t} = \cos(\omega t) \pm j \cdot \sin(\omega t)$$

■ Question 1 of 3 [8 marks]

Find the fundamental time period of the signals -

(a) [4 marks]

$$x[n] = 3.14 + 0.71 \cdot \sin\left(\frac{3\pi}{57}n + 33^\circ\right) + \cos\left(\frac{7\pi}{114}n - 16^\circ\right) - 0.24 \cdot \sin\left(\frac{7\pi}{114}n + 16^\circ\right)$$

(b) [4 marks]

$$x[n] = 0.43 + 0.45 \cdot \sin(0.45n + 30^\circ) + 0.16 \cdot \cos(0.15\pi n - 30^\circ)$$

■ Question 2 of 3 *[8 marks]*

Draw magnitude spectrum for the kernel $[-2, 0, \underline{4}, 0, -2]$.

The underlined value indicates the zero-th index. Maintain ranges for full marks.

■ Question 3 of 3 *[4 marks]*

Given you are applying Fourier analysis on the given time domain signal, what can you expect from the frequency spectrum? What would happen to the spectrum if the signal beyond the range $[-\pi, \pi]$ was made 0?

